

Jonas Lim Hong Xiang

Singapore | +65 9663 2152

✉ jonaslimhx@gmail.com | [LinkedIn](#) | [GitHub](#) | [Blog](#)

Generative AI and software engineer with 7 years building and operating production AI systems on AWS. Hands-on experience designing multi-agent LLM workflows, fine-tuning models, and deploying GenAI automation end-to-end – from identifying high-value opportunities through scalable, cost-efficient production. AWS Certified Solutions Architect (Professional) with a strong track record owning complex systems in regulated and compliance-aware environments.

PROFESSIONAL EXPERIENCE

Co-Founder & Software Architect

May 2024 - Jun 2026

NockHub, Singapore | nockhub.com

- Designed and deployed GenAI automation workflows on AWS serverless infrastructure (Lambda, API Gateway, DynamoDB), eliminating manual operational steps and reducing engineering overhead across the platform.
- Evaluated and integrated LLM tooling and multi-agent patterns into production pipelines; established model selection guidelines based on quality, latency, cost, and security requirements.
- Identified high-value automation opportunities by engaging stakeholders directly, balancing business impact against cost and scalability constraints.
- Defined engineering delivery standards and maintained production observability and cost controls across a team of 3 engineers and all AWS workstreams.
- Grew platform to 4 ranges, 7,000+ users, and 4,000+ bookings per month.

Senior Machine Learning Engineer

Jun 2022 - Jul 2024

Carro, Singapore

- Owned end-to-end delivery of production ML systems across predictive pricing, vehicle damage detection, and engine acoustics, deployed on AWS GPU infrastructure serving 12,000+ requests per day.
- Led model selection, fine-tuning, and production transition from Mask R-CNN to YOLOv5; reduced inference time from 500ms to 50ms through architecture and optimization iteration.
- Built a continuous data flywheel – data prep pipelines, feature store, and metadata store – improving training data quality by 50% and enabling faster, reproducible model improvement cycles.
- Implemented automated training, deployment, and monitoring pipelines; reduced operational overhead by 75 engineer-hours per month and improved production reliability.
- Optimized AWS infrastructure (networking, compute, databases); saved USD 120k in 2023 through cost and performance tuning.
- Partnered with product, operations, and data teams across Singapore, Malaysia, Indonesia, and Taiwan to translate business needs into AI system requirements.

Data Scientist

Jun 2019 - May 2022

Micron Technology, Singapore

- Built and deployed predictive ML models within a regulated semiconductor manufacturing environment with strict quality controls and compliance constraints; contributed to USD 11M in savings.
- Built a defect detection and classification model integrated into inspection tooling; improved tool capacity by 10%.
- Optimized SQL analytics pipelines for model training; reduced query time by 50% and improved data reliability for downstream ML workflows.

EDUCATION & AWARDS

Master of Computer Science (Data Science & AI)

Aug 2024 - Jun 2026

University of Sydney, Australia

Distinction | Singapore IMDA SG Digital Scholar

- Capstone: Designed and deployed a multi-agent Generative AI system on a Unitree Go2 quadruped robot – built LLM-driven agent workflows with model fine-tuning, tool use, task decomposition, and real-world autonomous execution.

Bachelor of Engineering (Aerospace)

Aug 2015 - May 2019

Nanyang Technological University, Singapore

First Class Honors, Highest Distinction

TECHNICAL EXPERTISE

- GenAI & LLM:** multi-agent systems, model fine-tuning, data flywheels, RAG, prompt engineering
- ML/AI:** PyTorch, TensorFlow, Hugging Face, embeddings, anomaly detection
- Languages:** Python, TypeScript, SQL, Dart, Bash
- Frontend / Mobile:** React, Flutter
- AWS:** Lambda, API Gateway, DynamoDB, S3, RDS, EC2, Terraform (AWS Certified Solutions Architect - Pro)
- Data & Analytics:** PySpark, SQL, ETL pipelines, feature stores, data quality
- Backend & APIs:** FastAPI, Docker, distributed systems
- Databases:** PostgreSQL, MySQL, MongoDB, Redis, DocumentDB

HACKATHONS

2025 Global Health Systems Hackathon Hub, "Building High-Value Health Systems: Leveraging AI". Team I Know You Know won 2nd place in Sydney and placed Top 20 globally.

CERTIFICATIONS

AWS Certified Solutions Architect - Professional

Jan 2024

Profiles:  [LinkedIn](#) |  [GitHub](#) |  [Blog](#)